



Constraints

Table Constraints :

- Constraints are rules that we set on database tables to maintain the accuracy and integrity of the data.
- They make sure the data entered into the table follows certain conditions so the database stays clean and reliable.

Types of Constraints:

1. PRIMARY KEY

- A Primary Key uniquely identifies each row (record) in a table.
- It cannot be empty (NOT NULL) and must be unique.
- Every table should have one primary key to identify records clearly.

Example:

StudentID in a Students table — no two students have the same ID.

2. FOREIGN KEY

- A Foreign Key links one table to another.
- It ensures the value in one table matches a value in another table's primary key.
- Helps maintain relationship between tables and keeps data consistent.

Example:

Linking StudentID in Enrollments table to Students table

3. UNIQUE

- The Unique constraint makes sure all values in a column are different.
- It allows no duplicates but can have NULL values (depending on the database).

Example:

Emails in a Users table must be unique so no two users have the same email.

4. NOT NULL

- The Not Null constraint makes sure a column cannot be empty.
- Data must be provided when inserting or updating rows.

Example:

A user's Name cannot be NULL because every user should have a name.

5. CHECK

- The Check constraint sets a rule that data in a column must satisfy.
- If the data doesn't meet the rule, the database rejects it.

Example:

Age must be greater than or equal to 18.

6. DEFAULT

- The Default constraint gives a column a default value if no value is provided.
- Useful to avoid empty fields and set common default info.

Example:

A Status column that defaults to 'Active' if no status is given.

Cosntraints.sql file:

```
CREATE TABLE Students (  
    StudentID INT PRIMARY KEY,      -- Unique ID for each student  
    Name VARCHAR(50) NOT NULL,      -- Name cannot be empty  
    Email VARCHAR(100) UNIQUE,      -- Email must be unique  
    Age INT CHECK (Age >= 18),      -- Age must be 18 or older  
    Status VARCHAR(10) DEFAULT 'Active' -- Default status is 'Active'  
);
```

-- Constraints

--Table: Employees

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,      -- PRIMARY KEY: unique, not null  
    DepartmentID INT,                -- FOREIGN KEY will be added later  
    Email VARCHAR(100) UNIQUE,       -- UNIQUE: no duplicate emails  
    Name VARCHAR(50) NOT NULL,       -- NOT NULL: name is required  
    Age INT CHECK (Age >= 18),       -- CHECK: age must be 18 or older  
    Status VARCHAR(10) DEFAULT 'Active' -- DEFAULT: default value is 'Active'  
);
```

-- Adding FOREIGN KEY constraint to link DepartmentID with Departments table

ALTER TABLE Employees

ADD CONSTRAINT FK_Department

FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID);

Note: For code practice follow my github repository copy and paste my code on your pc.

- https://github.com/HiraNawaz2415/SQL_practice1/blob/main/Constraints/02_constraints.sql
- https://github.com/HiraNawaz2415/SQL_practice1/tree/main